

B.Tech IV Year I Semester (R20) Regular & Supplementary Examinations December/January 2024/2025

INTRODUCTION TO COMPOSITE MATERIALS

(Common to CE, EEE, ECE, CSE, IT, FT, AI&DS, CSE (AI), CSE(DS), CSE(AI&ML), CSE(IOT), CSE(CS), CS&D and AI&ML)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
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| (a) Give some Fundamentals of Composite materials. | 2M |
| (b) What is glass Fiber? Explain. | 2M |
| (c) Explain about Filament winding. | 2M |
| (d) List the various applications of PMC'S. | 2M |
| (e) Differentiate between metals and composites. | 2M |
| (f) Write short note on Solid state Processes. | 2M |
| (g) Write short notes on Cold isostatic pressing. | 2M |
| (h) Explain about lanxide process. | 2M |
| (i) Write the composites used for aerospace applications. | 2M |
| (j) What are bio degradable composites? | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 Define composites. Classify them in accordance with the material and reinforcement types. 10M
- OR**
- 3 What is Fiber? Explain about Carbon and ceramic fibers with properties and applications. 10M
- 4 What is Filament winding? Give the various properties and applications of PMCs. 10M
- OR**
- 5 Explain about Injection molding and sheet molding compound of PCMs. 10M
- 6 Explain the various types of metal matrix composites. Write about the processing of MMC. 10M
- OR**
- 7 (a) What is the role of Reinforcements in Metal matrix Composites? 5M
(b) Explain the Advantages of MMCs over Polymer matrix Composites. 5M
- 8 Explain the applications of Ceramic matrix composites in detail. What are the advantages of Ceramic matrix composites over other composites? 10M
- OR**
- 9 (a) Briefly explain Liquid phase sintering with help of suitable examples. 5M
(b) What are In-situ chemical reaction techniques? 5M
- 10 (a) Write and explain about properties and applications of Carbon-carbon composites. 5M
(b) Give the limitations of carbon matrix and carbon fiber. 5M
- OR**
- 11 What are various applications of bio composites with respect to Mechanical, Biomedical, automobile Engineering? Explain in detail. 10M
